

# "Data Analytics Using Excel"

## Unit 1: Introduction to Data Analytics & Excel Basics

1. What is data analytics? Explain its importance and applications.
2. Describe the types of data analytics (Descriptive, Diagnostic, Predictive, Prescriptive).
3. Explain the key features of MS Excel useful for data analysis.
4. How to enter, format, and manage data in Excel?
5. Write the steps to create a worksheet and use Excel formulas.

## Unit 2: Excel Functions for Data Analysis

1. Explain the usage of \*Statistical Functions\* in Excel (AVERAGE, MEDIAN, MODE, STDEV).
2. Describe \*Logical Functions\* with examples (IF, AND, OR, NOT).
3. Write about \*Lookup & Reference Functions\* (VLOOKUP, HLOOKUP, MATCH, INDEX).
4. What are \*Text Functions\* in Excel? Give examples (LEFT, RIGHT, MID, CONCATENATE).
5. Explain \*Date and Time Functions\* with examples.

## Unit 3: Data Visualization in Excel

1. What are charts in Excel? Explain different types (Bar, Column, Pie, Line, etc.).
2. Explain the steps to insert and format a chart in Excel.
3. What is Conditional Formatting? How is it useful?
4. Describe the use of Sparklines and Data Bars.
5. Explain the importance of dashboards and how to build one in Excel.

## Unit 4: Data Tools in Excel

1. Explain the use of \*Sort and Filter\* options in Excel.
2. What is \*Data Validation\* in Excel? How is it applied?
3. Write about \*What-If Analysis\* (Goal Seek, Data Table, Scenario Manager).
4. Explain the purpose and steps to use \*Pivot Tables\*.
5. What are slicers in Excel? How do they help in data analysis?

## Unit 5: Case Studies and Practical Applications

1. Explain a case study where Excel is used for business decision-making.
2. Demonstrate how Excel can be used to analyze sales data.
3. Describe how to analyze student performance using Excel.
4. Prepare a sample budget report using Excel functions and formatting tools.
5. How to prepare and present data reports using Excel.